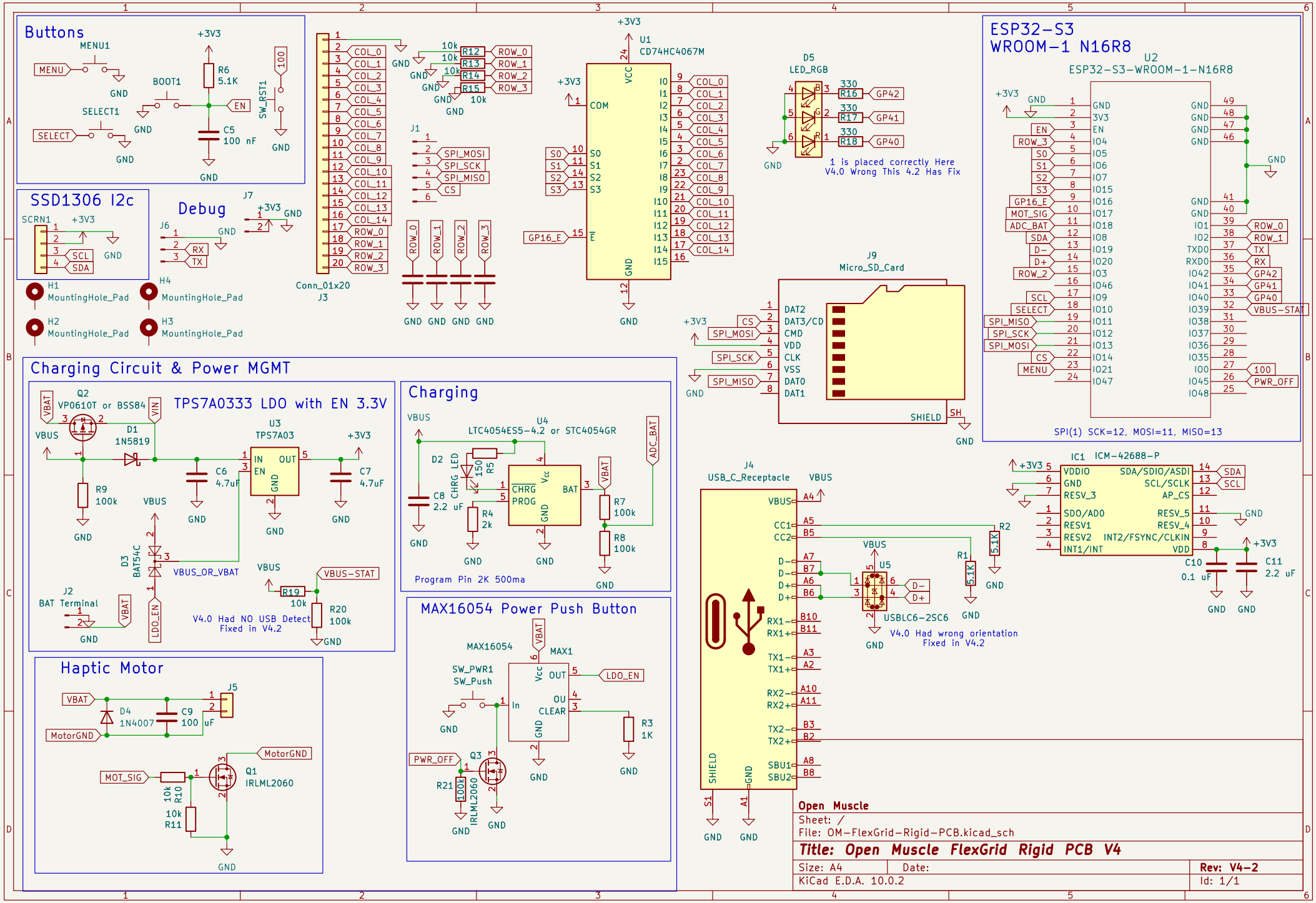

Buttons
MENU1
MENU
GND
BOOT1
SELECT1
SELECT
GND
GND
+3V3
R6
5.1K
EN
C5
100 nF
GND
SW_RST1
100
GND
SSD1306 I2c
SCRN1
+3V3
SCL
SDA
GND
Debug
J6
RX
TX
GND
J7
+3V3
GND
H1
MountingHole_Pad
H2
MountingHole_Pad
H4
MountingHole_Pad
H3
MountingHole_Pad
Conn_01x20
J3
COL_0
COL_1
COL_2
COL_3
COL_4
COL_5
COL_6
COL_7
COL_8
COL_9
COL_10
COL_11
COL_12
COL_13
COL_14
ROW_0
ROW_1
ROW_2
ROW_3
10k
R12
R13
R14
R15
J1
SPI_MOSI
SPI_SCK
SPI_MISO
CS
GND GND GND GND
+3V3
U1
CD74HC4067M
VCC
COM
S0
S1
S2
S3
EN
GND
GP16_E
I0
I1
I2
I3
I4
I5
I6
I7
I8
I9
I10
I11
I12
I13
I14
I15
D5
LED_RGB
330
R16
R17
R18
GP42
GP41
GP40
1 is placed correctly Here
V4.0 Wrong This 4.2 Has Fix
ESP32-S3
WROOM-1 N16R8
U2
ESP32-S3-WROOM-1-N16R8
SPI(1) SCK=12, MOSI=11, MISO=13
J9
Micro_SD_Card
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
SHIELD
SH
Charging Circuit & Power MGMT
TPS7A0333 LDO with EN 3.3V
Q2
VP0610T or BSS84
VBAT
VIN
VBUS
D1
1N5819
U3
TPS7A03
IN
OUT
EN
GND
C6
4.7uF
C7
4.7uF
R9
100k
D3
BAT54C
VBUS_OR_VBAT
LDO_EN
J2
BAT Terminal
V4.0 Had NO USB Detect
Fixed in V4.2
R19
10k
R20
100k
VBUS-STAT
Charging
U4
LTC4054ES5-4.2 or STC4054GR
D2
CHRG LED
150
R5
R4
2k
C8
2.2 uF
CHRG
PROG
Vcc
BAT
R7
100k
R8
100k
ADC_BAT
Program Pin 2K 500ma
MAX16054 Power Push Button
MAX16054
MAX1
SW_PWR1
SW_Push
Vcc
In
OUT
CLEAR
LDO_EN
R3
1K
PWR_OFF
Q3
IRLML2060
R21
100k
Haptic Motor
J5
D4
1N4007
C9
100 uF
MotorGND
MOT_SIG
Q1
IRLML2060
10k
R10
10k
R11
J4
USB_C_Receptacle
VBUS
CC1
CC2
D-
D+
RX1-
RX1+
TX1-
TX1+
RX2-
RX2+
TX2-
TX2+
SBU1
SBU2
SHIELD
R2
5.1K
R1
5.1K
U5
USBLC6-2SC6
V4.0 Had wrong orientation
Fixed in V4.2
IC1
ICM-42688-P
VDDIO
GND
RESV_3
SDO/AD0
RESV1
RESV2
INT1/INT
SDA/SDIO/ASDI
SCL/SCLK
AP_CS
RESV_5
RESV_4
INT2/FSYNC/CLKIN
VDD
C10
0.1 uF
C11
2.2 uF
Open Muscle
Sheet: /
File: OM-FlexGrid-Rigid-PCB.kicad_sch
Title: Open Muscle FlexGrid Rigid PCB V4
Size: A4
Date:
Rev: V4-2
KiCad E.D.A. 10.0.2
Id: 1/1